$Ax = b$

least squares.

$$A = \begin{pmatrix} 2 & 6 \\ 1 & -3 \\ 2 & 0 \end{pmatrix} \qquad b = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \qquad Ax = b \Rightarrow \begin{cases} 2x_1 + 6x_2 = 1 \\ x_1 - 3x_2 = 0 \\ 2x_1 = 0 \end{cases}$$

sistem incompatibil.

$\longrightarrow$ pseudosolutia:

$$\mathrm{Im}(A) = \mathrm{Sp}\left\{ \underbrace{\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 6 \\ -3 \\ 3 \end{pmatrix}}_{v_2} \right\} \xrightarrow[\text{procedeu}]{\text{Gram}}$$

$$\longrightarrow \left\{ g_1 = \frac{1}{3}\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, g_2 = \frac{1}{3}\begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix} \right\} \text{ bază ortonormată în } \mathrm{Im}(A)$$

$$\Rightarrow \underset{\mathrm{Im}(A)}{\mathrm{pr}}\, b = \underset{g_1}{\mathrm{pr}}\, b + \underset{g_2}{\mathrm{pr}}\, b = \langle b, g_1 \rangle g_1 + \langle b, g_2 \rangle g_2$$

$$= QQ^t b = \frac{1}{9}\begin{pmatrix} 8 \\ -2 \\ 2 \end{pmatrix}$$

$\operatorname{rang} A = 2 \Rightarrow A = QR$

$Q = (q_1 | q_2)$ matrice ortog., $R = \begin{pmatrix} 3 & 3 \\ 0 & 6 \end{pmatrix}$

$$\begin{cases} 2x_1^* + 6x_2^* = 8/9 \\ x_1^* - 3x_2^* = -2/9 \\ 2x_1^* = 2/9 \end{cases} \Rightarrow \begin{matrix} x_2^* = 1/9 \\ x_1^* = 1/9 \end{matrix} \quad \checkmark$$

$\Rightarrow$ pseudosol. sist. incompatibil este $x^* = \begin{pmatrix} 1/9 \\ 1/9 \end{pmatrix}$

$Ax = b$

$QRx = b \quad / \cdot Q^t$

$Q^t Q R x = Q^t b$

I

$Rx = Q^t b \quad / Q$

$Ax^* = QRx = QQ^t b = \operatorname{pr}_{\operatorname{Im} A} b$

$Rx^* = Q^t b \quad / R^{-1} \Rightarrow x^* = R^{-1} Q^t b$

R inversabilă triunghiulară

$(\operatorname{rang} A = n)$

$$\underbrace{Ax^*}_{\in \mathrm{Im}(A)} = \mathrm{pr}_{\mathrm{Im}(A)} b \iff A^t A x^* = A^t b$$

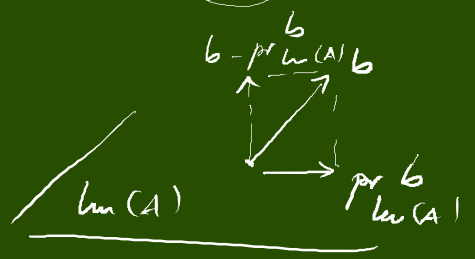
$$Ax^* = \mathrm{pr}_{\mathrm{Im}(A)} b \iff b - Ax^* \in \mathrm{Im}(A)^{\perp} =$$

$$= \mathrm{Ker}(A^t)$$

$$\iff A^t(b - Ax^*) = 0$$

$$\underbrace{A^t A}_{(n,m)(m,n)} x^* = A^t b$$

(n,n)



$A^t A$ inversible $\iff \mathrm{rang}\, A = n$

$A^t A$ inversible $\iff \mathrm{Ker}(A^t A) = \{0\}$

$\mathrm{Ker}(A^t A) = ?$

$x \in \mathbb{R}^n$. $x \in \mathrm{Ker}(A) \Rightarrow Ax = 0 \mid A^t$

$\Rightarrow A^t A x = 0 \Rightarrow x \in \mathrm{Ker}(A^t A)$

$\Rightarrow \quad \ker(A) \subseteq \ker(A^t A)$

$\supseteq$

$x \in \ker(A^t A) \Rightarrow A^t A x = 0 \;/\; x^t$

$(1,n)(n,m)(m,n)(n,1)$

$\Rightarrow \underbrace{x^t A^t}_{(Ax)^t} A x = 0 \qquad u^t u = \langle u, v \rangle$

$\Rightarrow \langle Ax, Ax \rangle = 0$

$\|Ax\|^2 = 0 \Rightarrow Ax = 0$

$\Rightarrow x \in \ker(A)$

$\Rightarrow \quad \ker(A) = \ker(A^t A)$

$A^t A$ invertible $\Leftrightarrow \ker(A^t A) = 0 \Leftrightarrow$

$\Leftrightarrow \ker(A) = 0$

$\dim_{\mathbb{R}} \ker(A) = n - \operatorname{rang}(A) \;/\; \Leftrightarrow$

$\Leftrightarrow \operatorname{rang}(A) = n \leq m$

Q: se A^tA non è invertibile

$A^tA x^* = A^t b$ non ammette sol. unica

$x^* \in \mathbb{R}^m$

- allora x^* di norma minima

$\min \|x^*\|^2$

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(\xi)}{2!}(x - x_0)^2$$

$f(x) \geq f(x_0) \quad \forall x$

$f''(\xi) > 0$

$$x^* = (A^tA)^{\dagger} A^t b$$

$A^\dagger$ pseudoinversa matrice A

dagger (inversa Moore-Penrose)

$Ax = b \qquad A = \begin{pmatrix} 2 & 6 \\ 1 & -3 \\ 2 & 0 \end{pmatrix} \qquad b = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$A^t A = \begin{pmatrix} 2 & 1 & 2 \\ 6 & -3 & 0 \end{pmatrix} \begin{pmatrix} 2 & 6 \\ 1 & -3 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 9 & 9 \\ 9 & 45 \end{pmatrix}$

$A^t A$ quadrada

$(A^t A)^t = A^t \cdot (A^t)^t = A^t A$

$\det A^t A = 9 \cdot 45 - 9 \cdot 9 = 9 \cdot 36 = 324$

$x^+ = \frac{1}{325} \begin{pmatrix} 45 & -9 \\ -9 & 9 \end{pmatrix} \cdot \underline{\begin{pmatrix} 2 & 6 \\ 1 & -3 \\ 2 & 0 \end{pmatrix}^t} \underline{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}$

$A^t A x^* = A^t b$

$\begin{cases} 9x_1^* + 9x_2^* = 2 \\ 9x_1^* + 45x_2^* = 6 \end{cases} \qquad 36x_2^* = 4 \quad x_2^* = 1/9$

$x_1^* = 1/9$

$y = \alpha e^{\beta x}$

$\ln y = \ln \alpha + \beta x$

$y = \alpha + \beta x + \gamma x^2$

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ $\quad X^2 + 1$ $\quad i \quad -i$

$A = A^t$ $\quad \exists\, \lambda \in \mathbb{C}$ (val. pr.) $\quad v \in \mathbb{C}^n$ vettore proprio $(v \neq 0)$

$Av = \lambda v$

$\bar{A}\bar{v} = \bar{\lambda}\bar{v} \Rightarrow A\bar{v} = \bar{\lambda}\bar{v}$

$A \in M_n(\mathbb{R}) \Rightarrow \bar{A} = A$

$\lambda \bar{v}^t v = \bar{v}^t (\lambda v) = \bar{v}^t (Av) = (\bar{v}^t A) v =$

$(1,n)(n,1)$

$= (\bar{v}^t A^t) v = (A\bar{v})^t v = \bar{\lambda} \bar{v}^t v$

$$v = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \neq 0 \qquad \bar{v}^t v = (\bar{z}_1 \dots \bar{z}_n)\begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}$$

$$= \bar{z}_1 z_1 + \dots + \bar{z}_n z_n$$

$$= |z_1|^2 + \dots + |z_n|^2 \neq 0$$

$$\lambda(\bar{v}^t v) = \bar{\lambda}(\bar{v}^t v) \Rightarrow \lambda = \bar{\lambda} \Rightarrow \lambda \in \mathbb{R}$$

Temă $A \in \mathcal{M}_n(\mathbb{R})$ $A^t = -A$ $\Rightarrow \sigma(A) = ?$

exemplu $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

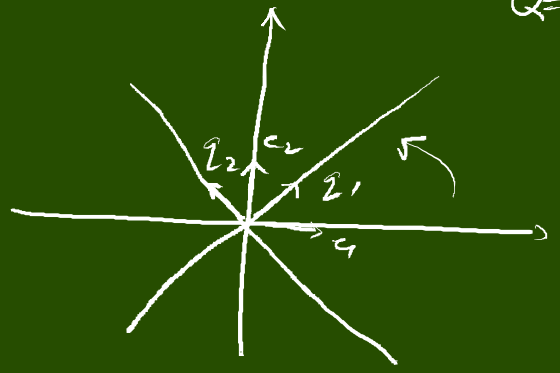
$$P_A(x) = x^2 - 4x + 3 \begin{cases} \lambda_1 = 3 \\ \lambda_2 = 1 \end{cases}$$

$\lambda_1 = 3$: $\begin{cases} 2x + y = 3x \\ x + 2y = 3y \end{cases} \Rightarrow x = y$

$V_{\lambda_1} = Sp\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\} \longrightarrow q_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$

$\lambda_2 = 1$: $\begin{cases} 2x+y = x \\ x+2y = y \end{cases} \Rightarrow x = -y$

$V_{\lambda_2} = \mathcal{L}\left\{\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\} \longrightarrow q_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$



$Q = (q_1 \mid q_2)$
= matrice ortog.

$Q^t Q = I_2$
(2,2)

$\Rightarrow A = Q \cdot D \cdot Q^{-1} = Q D Q^t$

$D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ ortodiagonalizzare

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$A = A^t \in M_n(\mathbb{R})$ $\lambda_1, \lambda_2 \in \sigma(A) (\subseteq \mathbb{R})$

$\lambda_1 \neq \lambda_2$

$v_1 \in V_{\lambda_1}$, $v_1 \neq 0$

$v_2 \in V_{\lambda_2}, \quad v_2 \neq 0 \quad \Rightarrow \quad v_1 \perp v_2$

$\langle v_1, v_2 \rangle = 0$

$\lambda_1 \langle v_1, v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle = \langle A v_1, v_2 \rangle =$

$= (A v_1)^t v_2 = v_1^t A^t v_2 = v_1^t A v_2 =$

$= v_1^t \lambda_2 v_2 = \lambda_2 \langle v_1, v_2 \rangle$

$\langle v_1, v_2 \rangle (\lambda_1 - \lambda_2) = 0 \quad \Rightarrow \quad \langle v_1, v_2 \rangle = 0$

$\neq 0$

$v_1 \perp v_2$

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esempio. $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ $\begin{matrix} 0 \\ 3 \end{matrix}$

$P_A(x) = \begin{vmatrix} 1-x & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix} = (3-x) \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix}$

$$= (3-x) \begin{vmatrix} 1 & 1 & 1 \\ 0 & -x & 0 \\ 0 & 0 & -x \end{vmatrix} = (3-x)\cdot x^2$$



$\lambda_1 = 3 \quad a(\lambda_1) = 1$

$\lambda_2 = 0 \quad a(\lambda_2) = 2$

$\lambda_1 = 3$:

$$\begin{cases} x+y+z = 3x \\ x+y+z = 3y \\ x+y+z = 3z \end{cases} \Rightarrow x=y=z$$

$$V_{\lambda_1} = Sp\left\{\underbrace{\begin{pmatrix}1\\1\\1\end{pmatrix}}_{v_1}\right\} \longrightarrow q_1 = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}$$

$\lambda_2 = 0$:

$$\begin{cases} x+y+z = 0 \\ x+y+z = 0 \\ x+y+z = 0 \end{cases}$$

$\Rightarrow z = -x-y \qquad \begin{pmatrix} x \\ y \\ -x-y \end{pmatrix}$

$$V_{\lambda_2} = Sp\left\{\underbrace{\begin{pmatrix}1\\0\\-1\end{pmatrix}}_{v_2}, \underbrace{\begin{pmatrix}0\\1\\-1\end{pmatrix}}_{v_3}\right\}$$

$$u_2 = v_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$u_3 = v_3 - \operatorname{pr}_{u_2} v_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1 \\ -1/2 \end{pmatrix}$$

$$q_2 = \frac{1}{\|u_2\|} u_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} =$$

$$q_3 = \frac{1}{\|u_3\|} u_3 = \sqrt{\frac{2}{3}}\begin{pmatrix} -1/2 \\ 1 \\ -1/2 \end{pmatrix} = \frac{1}{\sqrt{6}}\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} =$$

$\Rightarrow$ q_1, q_2, q_3 base ortonormată în $\mathbb{R}^3$

$$Q = (q_1 | q_2 | q_3) = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \end{pmatrix}$$

$$D = \begin{pmatrix} 3 & & \\ & 0 & \\ & & 0 \end{pmatrix}$$

$\Rightarrow$ $A = Q D Q^t$

I_3 $\quad \lambda_1 = 1 \quad a(\lambda_1) = 3$

$$aI_3 + A = aI_3 + \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}}_{A} \quad \begin{matrix} 3 \\ 0,0 \end{matrix}$$

$= P(A)$

$$= \begin{pmatrix} a+1-1-1 & 1 & 1 \\ 1 & a+1-1-1 & 1 \\ 1 & 1 & a+1-1-1 \end{pmatrix} \quad + a_1 + a_2 3 + a$$

$A \longrightarrow \lambda \qquad P(A) \longrightarrow P(\lambda)$

$P(x) = a + x$

$P(3) = a + 3$

$P(0) = a \; \checkmark \longrightarrow \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$

$$A = \underbrace{\lambda_1 \underbrace{q_1 q_1^t}_{\text{rang } 1}}_{(n,1)(1,n)} + \ldots + \lambda_n \underbrace{q_n q_n^t}_{\text{rang } 1}$$

(n,n)

$P_i = q_i q_i^t \in \mathcal{M}_n(\mathbb{R})$

$\operatorname{rang} P_i = 1$

$P_i^2 = q_i (q_i^t q_i) q_i^t = q_i q_i^t = P_i$

$q_i^t q_i = 1$

proiector pe $\mathcal{L}\{q_i\}$

$i \neq j$ $\quad P_i P_j = q_i (q_i^t q_j) q_j^t = 0 = P_j P_i$

$q_i^t q_j = 0$

$(P_i + P_j)^2 = P_i^2 + P_j^2 = P_i + P_j$

proiector pe $\mathcal{L}\{q_i, q_j\}$

$A = \sum_{i=1}^{n} \lambda_i P_i$

$\sum_{i=1}^{n} P_i = I_n$

$A = A^t$ $\quad A = Q \cdot \Delta \cdot Q^t$

$Q = (q_1 | \dots | q_n)$ vect. ortonormați

$$D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$\lambda_1 \geq \dots \geq \lambda_n$.

i) $\max_{\substack{x \in \mathbb{R}^n \\ \|x\| = 1}} x^t A x = ?$ $\qquad x_M = ?$

ii) $\min_{\substack{x \in \mathbb{R}^n \\ \|x\| = 1}} x^t A x = ?$ $\qquad x_m = ?$

iii) $\max_{\substack{x \in \mathbb{R}^n \\ \|x\| = 1 \\ x \perp q_1}} x^t A x = ?$ $\qquad x_0 = ?$